

Tenacious-Bench v0.1 — Final Evaluation Memo

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Page 1 — The Decision

Executive Summary

A CPO-trained Qwen2.5 0.5B judge, fine-tuned on 137 Tenacious-Bench preference pairs, correctly classifies **92.7%** of held-out compliance tasks (51/55) versus **69.1%** for the rule-only baseline — a **+23.6 pp gain (95% CI: [+9.8, +37.4], McNemar p=0.004)** — while adding zero marginal cost at inference. The trained judge closes three of four documented failure modes (honesty flag bypass, bench overcommitment, tone drift) but still requires the rule-based layer for scheduling and format checks; the recommended deployment is a hybrid: rule-only as a hard gate, CPO judge as the semantic layer, with a human-review trigger for low-confidence outputs.

Headline Lift — Delta A

Condition	Held-out Accuracy	Pass Rate	Cost / task
Rule-only baseline	69.1% (38/55)	61.8%	\$0.000
Prompt-eng (3-shot Haiku)	78.2% (43/55)	65.5%	\$0.001
CPO judge v1 (trained)	92.7% (51/55)	67.3%	\$0.000
Claude Sonnet API (est.)	~96.0%	~68%	\$0.003

Delta A = +23.6 pp over rule-only (95% CI: [+9.8 pp, +37.4 pp], McNemar $\chi^2=8.47$, p=0.004). Largest per-category gains: multithread-leakage +70 pp, cost-pathology +50 pp, dual-control +38 pp.

Delta B — Honest Report

Delta B = +14.5 pp over 3-shot prompt-engineering (95% CI: [+1.6 pp, +27.4 pp], McNemar p=0.043). The lower bound (+1.6 pp) is near zero — a stronger few-shot baseline would narrow this gap. The advantage of the trained judge is not accuracy alone: it costs **\$0.000/task** versus \$0.001/task for prompt-engineering and was trained on the exact violation types in the Tenacious domain.

Cost Per Task

Mode	Cost/task	Cost/1k tasks	Notes
Rule-only	\$0.000	\$0.00	No API call
CPO judge (local 4-bit)	\$0.000	\$0.00	Runs on CPU after quantization
Prompt-eng (Haiku)	\$0.001	\$0.80	3-shot, ~800 tokens/call
Claude Sonnet judge	\$0.003	\$3.30	Full eval-tier, not deployed

Deployment Recommendation

Deploy with caveat. Deploy the hybrid system: (1) rule-only hard gate for format/length/banned-phrase violations, (2) CPO judge for semantic compliance (honesty flags, signal grounding), (3) human-review queue for judge scores in [0.60–0.75]. Caveat: the judge was trained on 137 pairs — half the minimum recommended by Liu et al. (2024) for this task complexity. ICP misclassification accuracy (96%) masks insufficient coverage of diverse disqualification scenarios. Deploy on four high-confidence categories first (signal-overclaiming, tone-drift, bench-overcommitment, dual-control); gate ICP misclassification behind human review until training data doubles.

Page 2 — The Skeptic's Appendix

Four Failure Modes v0.1 Does Not Capture

- Multi-turn context degradation.** All 274 tasks are single-turn (brief → email). Agents operating in a thread context accumulate prior turns that can override active honesty flags. *v0.2 addition:* multi-turn trace tasks with session-state honesty flags.
- Cross-prospect data leakage.** When an agent processes briefs for multiple prospects in a batch, signals from prospect A can leak into prospect B's email. No v0.1 task tests for cross-contamination. *v0.2 addition:* paired tasks where the second task contains a signal that should only be present in the first.
- Partial flag acknowledgment.** The rubric marks FAIL if any banned pattern from an active flag appears. A finer failure mode is partial acknowledgment: agent hedges one flag correctly while ignoring a second. The binary FAIL verdict does not distinguish "ignored all flags" from

"ignored one of three flags." *v0.2 addition:* per-flag scoring with a minimum-satisfied-flags threshold.

4. Prospect response simulation. The benchmark evaluates the outbound email in isolation and does not assess whether a compliant email elicits a positive response. A score-1.0 email can still be a weak opener. *v0.2 addition:* a prospect-response simulator that rates conversion likelihood independently of compliance.

Public-Signal Lossiness in Ground Truth

Hiring signals, AI maturity scores, and leadership signals are synthetic parameters drawn from public data *schemas* (Crunchbase field structures, LinkedIn job post patterns) but not from live scrapes. The benchmark does not test signal staleness (a hiring signal true six months ago but no longer active), and the AI maturity score is a numeric placeholder, not a real inferred score from a production signal pipeline. Ground truth labels are reliable for the rubric dimensions as written; the rubric itself does not capture signal temporality. Any agent that correctly handles synthetic signals still needs separate evaluation on live-signal inputs before production deployment.

One Honest Unresolved Training Failure

The CPO judge has a **PASS bias on ICP misclassification tasks**. Tasks where the prospect is disqualified but the email looks compliant (no banned phrases, correct format, generic but not lying) score 1.0 on three of four rubric dimensions before the judge is applied. The judge learns surface compliance correctly but cannot reliably detect the policy-level suppression decision (this prospect should receive no email at all) from the email text alone. ICP misclassification accuracy is 96% in ablations — but this largely reflects structural email properties, not the suppression decision. This failure mode is unresolved and requires a separate ICP classifier gate upstream of the email judge.

Kill-Switch Trigger Condition

Trigger the kill-switch and revert to rule-only if any of the following are observed in production:

1. The judge's false-negative rate on **layoff_overrides_funding violations exceeds 5%** over a 7-day rolling window (measured via weekly audit of 50 randomly sampled judge-approved emails). A false negative sends a growth pitch to a company in active workforce reduction — direct reputational risk.
2. The judge approves any email containing a **specific banned phrase from Style Guide v2** that also appears in a tone-drift violation documented in the training data. This indicates regression on a learned pattern.
3. **Classification accuracy on the dev partition drops below 70%** after any model update or quantization change — measured by running `scoring_evaluator.py --batch tenacious_bench_v0.1/dev/ --no-llm` on merged weights before deployment.

All numeric claims mapped to source files in evidence_graph.json. Full methodology: methodology_rationale.md. Training artifacts: training/. Ablations: ablations/.